Polyatomic Ions

CHEMISTRY REFERENCE

★ **Memorize these 15.** The rest are here to look up. Each group lists the ions to memorize first.

1+ ions

★	**ammonium**	**NH_4^+**
★	**hydronium**	**H_3O^+**

2+ ions

	mercury(I)	Hg_2^{2+}

1– ions

★	**acetate**	**$C_2H_3O_2^-$**
★	**hydrogen carbonate (bicarbonate)**	**HCO_3^-**
★	**chlorate**	**ClO_3^-**
★	**cyanide**	**CN^-**
★	**hydroxide**	**OH^-**
★	**nitrate**	**NO_3^-**
★	**nitrite**	**NO_2^-**
★	**permanganate**	**MnO_4^-**
	perchlorate	ClO_4^-
	chlorite	ClO_2^-
	hypochlorite	ClO^-
	bromate	BrO_3^-
	iodate	IO_3^-
	periodate	IO_4^-
	hydrogen sulfate (bisulfate)	HSO_4^-
	hydrogen sulfite (bisulfite)	HSO_3^-
	hydrogen sulfide (bisulfide)	HS^-
	dihydrogen phosphate	$H_2PO_4^-$
	thiocyanate	SCN^-
	cyanate	OCN^-

2– ions

★	**carbonate**	**CO_3^{2-}**
★	**chromate**	**CrO_4^{2-}**
★	**sulfate**	**SO_4^{2-}**
★	**sulfite**	**SO_3^{2-}**
	dichromate	$Cr_2O_7^{2-}$
	hydrogen phosphate	HPO_4^{2-}
	oxalate	$C_2O_4^{2-}$
	peroxide	O_2^{2-}
	thiosulfate	$S_2O_3^{2-}$
	silicate	SiO_3^{2-}

3– ions

★	**phosphate**	**PO_4^{3-}**
	phosphite	PO_3^{3-}
	arsenate	AsO_4^{3-}
	borate	BO_3^{3-}